Subject: Mathematics | **CBSE CLASS 10 – WEEKLY TEST 06** | **Time:** 45 Minutes
Class: Class X (CBSE) | **Chapter 13: Statistics** | **MM: 20**

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Internal choices are provided in question subparts.
2. **Section A** comprises 4 MCQs and 1 Assertion-Reason question of 1 mark each.
3. **Section B** comprises 1 Very Short Answer (VSA) question of 2 marks.
4. **Section C** comprises 3 Short Answer (SA) questions of 3 marks each.
5. **Section D** comprises 1 Case Study based question of 4 marks with internal subparts.
6. Use of calculators is not permitted. All rough work and calculation steps must be clearly shown.

SECTION A: Multiple Choice & Assertion-Reason (5 × 1 = 5 Marks)

Q1. If the median of a distribution is 20 and each observation is increased by 2, then the new median will be: [1]

(A) 20 (B) 22
(C) 40 (D) Cannot be determined

Q2. The relationship between mean, median, and mode for a moderately skewed distribution is: [1]

(A) Mode = 2 Median − 3 Mean
(B) Mode = 3 Median − 2 Mean
(C) Median = 3 Mode − 2 Mean
(D) Mean = 3 Median − 2 Mode

Q3. If the mode of a data is 45 and the mean is 27, then its median is: [1]

(A) 30 (B) 33 (C) 36 (D) 32

Q4. For the following data, the lower limit of the modal class is: [1]

Age (years)	0–10	10–20	20–30	30–40	40–50
Patients	6	11	21	23	14

(A) 20 (B) 30 (C) 23 (D) 40

Q5. **Assertion (A):** If the mean and median of a distribution are 15 and 16 respectively, then its mode is 18.
Reason (R): Empirical formula connecting central tendencies is Mode = 3 Median − 2 Mean. [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(C) (A) is true but (R) is false. (D) (A) is false but (R) is true.

SECTION B: Very Short Answer Type (1 × 2 = 2 Marks)

Q6. Find the median class and modal class for the following distribution: [2]

Class Interval	0–10	10–20	20–30	30–40	40–50
Frequency	3	8	15	7	5

SECTION C: Short Answer Type (3 × 3 = 9 Marks)

Q7. The mode of the following distribution is 36. Find the missing frequency x: [3]

Class	0–10	10–20	20–30	30–40	40–50	50–60
Frequency	5	8	x	16	12	6

Q8. Find the mean of the following distribution by the Step-Deviation Method: [3]

Class Interval	50–52	53–55	56–58	59–61	62–64
Frequency	15	110	135	115	25

Q9. Find the median of the following frequency distribution of examination marks: [3]

Marks	15–25	25–35	35–45	45–55	55–65	65–75
Frequency	8	10	15	25	40	20

SECTION D: Case Study Based Question (1 × 4 = 4 Marks)

Q10. CASE STUDY: COMMUNITY CLUB TAMBOLA GAME NUMBER DRAW FREQUENCY [4 Marks]

At an annual community festival, a game of Tambola (Housie) with numbered tokens from 1 to 90 was organized. The caller recorded the frequency distribution of numbers drawn across 80 completed rounds.

Number Range	1–15	16–30	31–45	46–60	61–75	76–90
Draw Frequency	8	14	22	18	12	6

(i) State the class interval showing the maximum draw frequency. [1]

(ii) Convert the intervals into continuous true limits and compute the mode. [2]

(iii) Find the cumulative frequency of tokens drawn with value 60 or less. [1]

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★★★ END OF QUESTION PAPER · ALL THE BEST ★★★